

TikZ Crash Course

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TikZ is a LaTeX package for drawing directly in a document. Pandoc passes raw `\begin{tikzpicture}...\end{tikzpicture}` blocks through unchanged to xelatex, so every diagram below is a real vector drawing, not an image, and needs no external tools to build.

Every TikZ picture is a list of drawing commands ended with a semicolon, inside a `tikzpicture` environment. The two commands used most are `\draw` (paths: lines, shapes, curves) and `\node` (a labeled box, circle, or point you can refer to by name).

1. Coordinates and paths

`\draw` connects coordinates with `--` for straight lines. Shapes like `circle` and `rectangle` are keywords used after a coordinate.

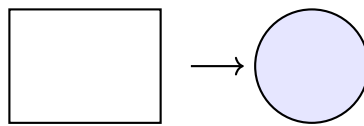


Figure 1: A rectangle, an arrow, and a filled circle

2. Nodes

A **node** is a piece of content (usually text) with an optional shape drawn around it. Naming a node with `(name)` lets later commands refer back to its position.



Figure 2: Two independent nodes, placed by explicit coordinates

3. Connecting nodes into a flowchart

Once nodes are named, `\draw (a) -- (b)` connects them without recomputing coordinates. Adding `->` (or the sharper `-Stealth`, from the `arrows.meta` library) turns a line into an arrow, which is how most flowcharts are built.



Figure 3: A simple left-to-right flowchart

4. Relative positioning

Explicit coordinates get tedious for bigger diagrams. The `positioning` library adds `below=of`, `right=of`, etc., so a node's location is described relative to another node instead of measured by hand.

5. Repetition with `\foreach`

`\foreach` loops over a list of values, running the same drawing command for each one. This avoids writing near-identical `\draw` lines by hand.

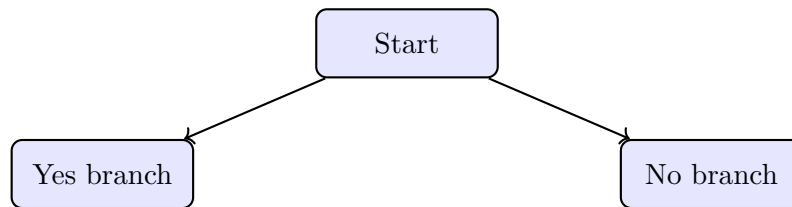


Figure 4: Nodes placed relative to each other, not by coordinates

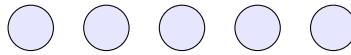


Figure 5: Five circles drawn from one loop instead of five `\draw` lines

Where to go from here

These five patterns, coordinates, nodes, connections, relative positioning, and loops, cover most flowcharts, diagrams, and simple plots you'll want in a document. The full TikZ & PGF manual documents many more shapes, libraries, and styling options.